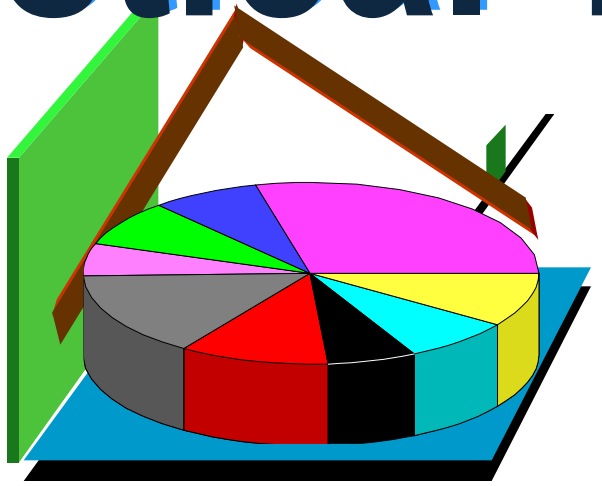
AA 200 - BIOMETRY

EXPLANATION OF Statistical Terms



STATISTICS

Is a set of scientific principles and techniques that are useful in reaching conclusions about a population and processes when available information is both limited and variable – *science of learning from data*

DESCRIPTIVE STATISTICS

Involves the arrangement, summary, and presentation of data, to enable meaningful interpretation, and to support decision making.

INFERENTIAL STATISTICS

Branch of statistics that focuses on using sample data to make generalizations, predictions, or inferences about a larger population.

INFERENTIAL STATISTICS

Unlike descriptive statistics, which summarize and describe the characteristics of a dataset, inferential statistics aim to draw conclusions that extend beyond the immediate data.

INFERENTIAL STATISTICS

This is particularly useful when it is impractical or impossible to collect data from an entire population.

IMPORTANCE OF INFERENTIAL STATISTICS

1. Generalization: Allows researchers to make broad conclusions from limited data.

2. Decision-Making: Provides a basis for making informed decisions in fields like medicine, agriculture, economics, and social sciences.

3. Prediction: Enables the prediction of future trends or outcomes based on current data.

POPULATION

Entire set of individuals, items, or data points that are of interest in a particular study.

It includes every possible member of the group being studied.

CHARACTERISTICS OF A POPULATION

1. Complete group from which you want to draw conclusions.
2. It can be finite (e.g., all students in a school) or infinite (e.g., all possible outcomes of a repeated experiment).
3. Often too large or impractical to study in their entirety.

SAMPLE

1. Subset of the population that is selected for analysis.

2. Used to represent the population and make inferences about it.

CHARACTERISTICS OF A SAMPLE

Aspect	Population	Sample
Definition	The entire group of interest.	A subset of the population.
Size	Can be very large or infinite.	Smaller and manageable.
Purpose	To describe or analyze the whole group.	To make inferences about the population.
Feasibility	Often impractical to study entirely.	Practical and cost-effective to study.
Example	All students in a university.	200 students randomly selected from the university.

WHY USE A SAMPLE

Practicality: It is often impossible or too costly to collect data from an entire population.

Efficiency: Samples allow researchers to gather data quickly and efficiently.

Accuracy: With proper sampling techniques, a sample can provide highly accurate estimates of population parameter

STATISTIC

A numerical characteristic of a sample (e.g., sample mean, sample proportion).

PARAMETER

A numerical characteristic of a population (e.g., population mean, population proportion).

VARIABLE

A variable - a characteristic of population or sample that is of interest for us.

1. Cereal choice
2. Capital expenditure
3. The waiting time for medical services

DATA

Data - the actual values of variables

1. **Quantitative data are numerical observations**
2. **Qualitative data are categorical observations**

Types of data

1. A variable - a characteristic of population or sample that is of interest for us.
    - Cereal choice
    - Capital expenditure
    - The waiting time for medical services

2. Data - the actual values of variables
    - Quantitative data are numerical observations
    - Qualitative data are categorical observations

PARAMETRIC AND NON-PARAMETRIC STATISTICS

1. **Parametric** and **non-parametric statistics** are two broad categories of statistical methods used for data analysis.

2. The choice between them depends on the nature of the data and the assumptions that can be made about the population from which the sample (data) is drawn.

PARAMETRIC STATISTICS

Key Assumptions

Normality: The data is normally distributed (or approximately normal).

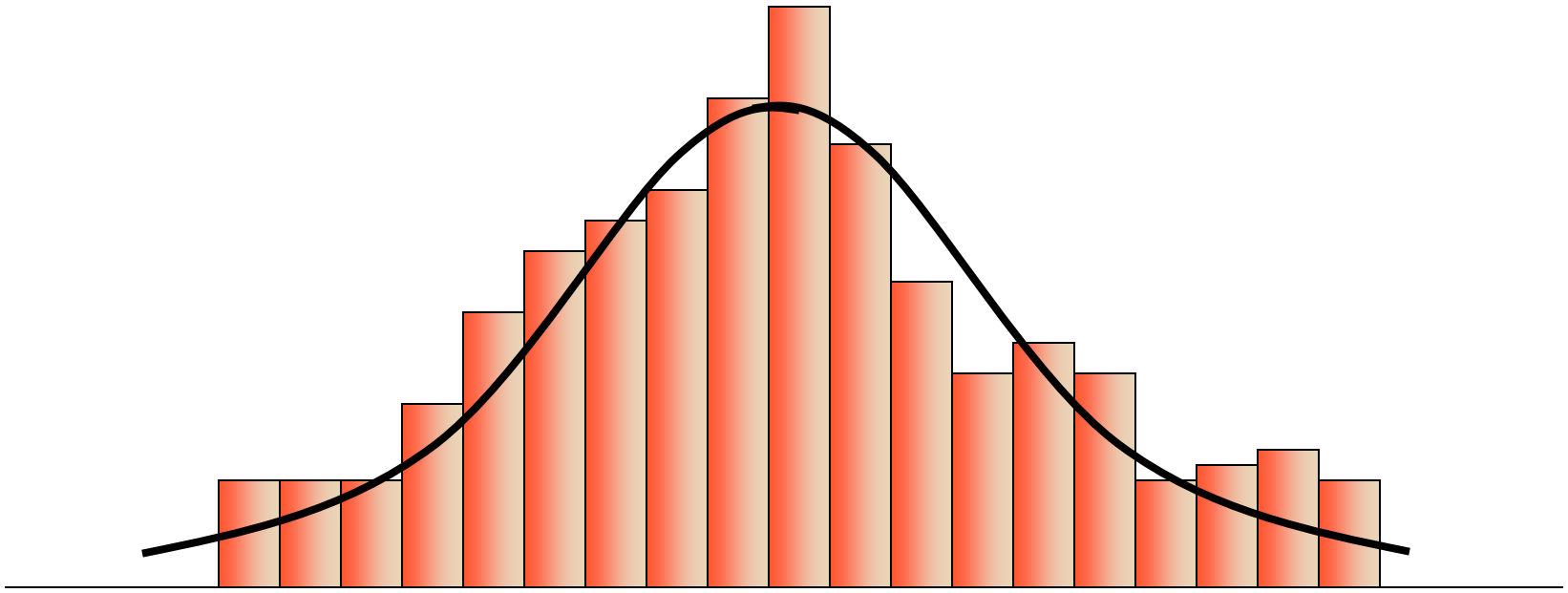
Homogeneity of Variance: The variance within groups is similar (for tests comparing groups).

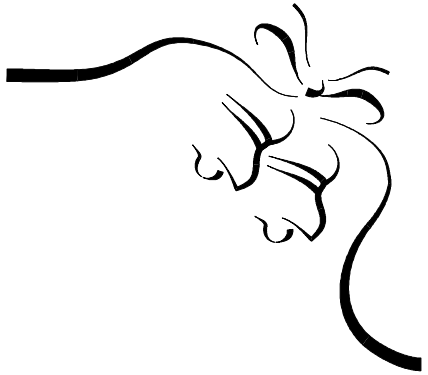
Interval or Ratio Scale: The data is measured on an interval or ratio scale (e.g., temperature, height).

Independence: Observations are independent of each other.

Bell shaped histogram

- Many statistical techniques require that the population be bell shaped.
- Drawing the histogram helps verify the shape of the population in question





Skewness



Negatively skewed

Positively skewed

PARAMETRIC STATISTICS

Advantages

1. More **statistical power** (higher likelihood of detecting true effects).
2. Can provide more **precise estimates** when assumptions are met.

Disadvantages

1. Sensitive to violations of assumptions (e.g., non-normality, unequal variances).
2. Not suitable for small sample sizes or non-normal data.

NON-PARAMETRIC STATISTICS

1. Non-parametric statistics are methods that do not rely on assumptions about the population distribution or parameters.

2. They are often referred to as **distribution-free** tests and are used when data does not meet the assumptions required for parametric tests.

CHARACTERISTICS OF NON-PARAMETRIC STATISTICS

1. **No Assumption of Normality**: Suitable for non-normal or skewed data.

2. **Ordinal or Nominal Scale**: Can be used with data measured on ordinal (e.g., rankings) or nominal (e.g., categories) scales.

3. **Small Sample Sizes**: Often used when sample sizes are small.

4. **Robustness**: Less sensitive to outliers and extreme values.

CHARACTERISTICS OF NON-PARAMETRIC STATISTICS

Advantages

1. Flexible: Can be used with data that does not meet parametric assumptions.

2. Robust: Less affected by outliers or non-normal distributions.

3. Applicable to Small Samples: Suitable for small or unevenly distributed datasets.

Disadvantages

1. Less Statistical Power: May fail to detect true effects when compared to parametric tests.

2. Less Informative: Do not provide estimates of population parameters (e.g., mean, variance).

DIFFERENCES BETWEEN PARAMETRIC AND NON-PARAMETRIC STATISTICS

Aspect	Parametric Statistics	Non-Parametric Statistics
Assumptions	Assumes normality, homogeneity, etc.	No assumptions about population distribution.
Data Scale	Interval or ratio scale.	Ordinal, nominal, or non-normal data.
Statistical Power	Higher power when assumptions are met.	Lower power compared to parametric tests.
Sensitivity	Sensitive to outliers and violations.	Robust to outliers and non-normality.
Examples	T-tests, ANOVA, Pearson correlation.	Mann-Whitney U, Wilcoxon, Chi-square.

NEXT LESSON – PRIVATE PREPARATION

1. TYPES OF VARIABLES AND DATA

2. SCALES OF MEASUREMENT

a) NOMINAL
b) ORDINAL
c) INTERVAL
d) RATIO

Thank You

